MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG4DSCECT202 - IOT System Design

Practical Examination

Duration: 2 hours Maximum Marks: 35

Instructions:

Demonstrate the given task (Any one of the tasks listed below)

Viva session: (Minimum 5 questions from the practical mode)

Lab Report

Evaluation Criteria:

- Viva: 10 marks
- Lab Report: 10 marks
- Demonstration: 15 marks

Tasks: (Select any 1 Exercise:)

1. **Write and execute a program to blink an LED using ESP8266**

2.Design a circuit and write a program to control an LED using a push button switch using ESP8266

3.Interface a PIR sensor with ESP8266 and generate an alert using the Blynk application.

4. **Interface an ultrasonic sensor with ESP8266 to measure distance.**

5. **Interface an infrared (IR) obstacle sensor with ESP8266 and generate an alert using the Blynk app.**

6.Interface an LM35 temperature sensor with ESP8266.
Read the temperature and display the measurement on the serial monitor.

7.Interface a DHT11 sensor with ESP8266 and display temperature and humidity values on the serial monitor.

8. **Interface a soil moisture sensor with ESP8266 and display soil moisture level on the serial monitor.**

9.Interface a rain drop sensor with ESP8266 and display rain detection status.

10.Control a servo motor using the Blynk application and ESP8266.

11.Generate a PWM signal using ESP8266 and observe the output using a CRO.

12.Implement brightness control of an LED using PWM technique.

13.Interface a DC motor with ESP8266 and control its operation using the Blynk application.

14. **Interface an LCD display with ESP8266 and display appropriate messages.**

15. **Interface LM35 temperature sensor with ESP8266 and IoT platform (Blynk).**
Display temperature values on the Blynk dashboard.

16.Interface DHT11 sensor with ESP8266 and IoT platform (Blynk).
Display temperature and humidity on the Blynk mobile application.

17. **Interface a soil moisture sensor with ESP8266 and IoT platform (Blynk).**
Monitor soil condition remotely.

18. **Interface a rain drop sensor with ESP8266 and IoT platform (Blynk).**
Display rain status on the mobile application.

19. **Design and implement an IoT-based home appliance control system using relay and Blynk application.**
Demonstrate remote ON/OFF control of appliances.